14	**(a)**		If $n = 1, n + 1 = 2$ and $n^2 + n + 1 = 3$	**E1** **[1]**	**2.2a**		
14	**(b)**		For n a positive integer, $\frac{1}{n}\left(\frac{1}{n^2+n+1}\right) < 1$ $arctan\left(\frac{1}{n+1}\right) + arctan\left(\frac{1}{n^2+n+1}\right)$ $= arctan\left(\frac{\left(\frac{1}{n+1}\right)+\left(\frac{1}{n^2+n+1}\right)}{1-\left(\frac{1}{n+1}\right)\left(\frac{1}{n^2+n+1}\right)}\right)$ $arctan\left(\frac{n^2+n+1+n+1}{(n^2+n+1)(n+1)-1}\right)$ $\arctan\left(\frac{n^2+2n+2}{n^3+2n^2+2n}\right) = \arctan\left(\frac{n^2+2n+2}{n(n^2+2n+2)}\right)$ $= \arctan\left(\frac{1}{n}\right)$	**E1** **M1** **M1** **A1** **[4]**	**2.3** **3.1a** **1.1** **2.1**	Use of arctan addition formula Clearing fractions within the fraction Condone omission of arctan for this mark and condone 1 other error Convincing completion (**AG**)	